FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2019

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABCD

Sol. The above lens mirror combination is an equivalent converging mirror.

2. ACD

Sol. $PV=nRT$ ---------- (1)

Mass of piston is 9kg & area of piston is $0.09m^2$

So

$$P=P_0+\frac{9g}{0.09}$$

$$P=1.06\times10^5 N/m^2$$

As mass of piston is not changed & Piston is frictionless so process is isobaric

$$P_f=P_i=1.06\times10^5 N/m^2$$

$$P_f=P_i=1.06\times10^5 N/m^2$$

$$dQ=n\,Cp\,dT\text{----------(2)}$$

$$\text{or } P_2V_2-P_1V_1=nR\,dT\text{-------(3)}$$

$$(3)\div(2) \qquad \frac{P(V_2-V_1)}{dQ}=\frac{2}{5}$$

$$\Rightarrow P(V_2-V_1)=\frac{2}{5}\times2.5\times10^4$$

$$W=P\Delta V=10^4 J \quad \& \text{ by equation } (1)$$

$$n=\frac{1.06\times10^5\times0.0027}{R\times300}$$

$$n=\frac{1.06\times270}{2500}=0.11448$$

or by $dQ=n\,Cp\,dT$

$$\therefore T_2-T_1=\frac{2.5\times10^4}{0.11448\times\frac{5R}{2}}$$

on solving $T_2=1.0782.18K$

3. AC

Sol. $f=mg\sin\theta$ and $PE\ \sin\theta=mg\sin\theta.R$

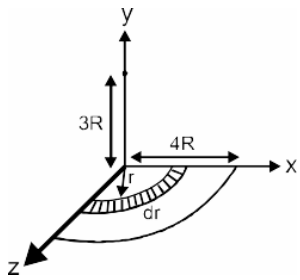
$q\ 2\ R\ E\sin\theta=mg\sin\theta\ R$

$$E=\frac{mg}{2q}$$

4. ABCD

Sol. (A) $$v=\int_0^{4r}\frac{K\frac{\pi}{2}rdr\sigma}{\sqrt{r^2+(3R)^2}}=\frac{\sigma R}{4\varepsilon_0}$$

(B) $$v=\int_0^{4r}\frac{K\frac{\pi}{2}rdr\sigma}{r}=\frac{\sigma R}{2\varepsilon_0}$$



5. BD

Sol. External electric field and induced charge on outer surface of the sphere will produce a net zero field inside it. Net field at P must be vertically upward.

6. ABCD

Sol. At constant temperature, decrease in molecular mass causes flattening of the graph. For same molecular mass of gas, increase in temperature causes flattening of the graph.

$$\left(V_{mp}\right)_{T_1}<\left(V_{mp}\right)_{T_2}$$

$$\therefore\frac{T_1}{M_A}<\frac{T_2}{M_B}\qquad\therefore\frac{T_2}{T_1}>\frac{M_B}{M_A}$$

But, $\frac{M_B}{M_A}$ can be less than or greater than 1

Similarly, $\frac{M_A}{M_B}>\frac{T_1}{T_2}$

If gas A is O_2 and gas B is N_2, then $M_A>M_B$.

$$\left(V_{mp}\right)_{T_1}<\left(V_{mp}\right)_{T_2}$$

$\therefore \frac{T_1}{M_A} < \frac{T_2}{M_B}$ $\quad\quad$ $\therefore \frac{T_2}{T_1} > \frac{M_B}{M_A}$

7. D
8. A

Sol. **(Sol. for Q. 7 and Q.8)**

Let the initial temperature, pressure and volume of gas in 'A' be T_0, P_0, V_0 and the area of the position A and B be a, 2a. Now gas in chamber 'A' undergoes adiabatic compression whereas gas in chamber 'B' undergoes isothermal expansion.

Now solving for gas in 'A'

$P_{1A}V_{1A}^{\gamma} = P_{2A}V_{2A}^{\gamma} \Rightarrow P_0V_0^{\gamma} = P_{2A}(0.25V_0) \Rightarrow P_{2A} = 8P_0$

Solving for gas in 'B'

$P_{1B}V_{1B} = P_{2B}V_{2B} \Rightarrow P_{1B}V_0 = P_{2B} \times 2.5V_0 \Rightarrow P_{2B} = 0.4P_{1B}$

Also $(P_{2A} \times a) = (P_{2B} \times 2a)$ for gas is chambers 'B'

$\Rightarrow 8P_0a = P_2 \times 2a \Rightarrow P_2 = 4P_0$

$\Rightarrow 0.4P_1 = 4P_0 \Rightarrow P_1 = 10P_0$

Now comparing the moles of gas in A and B

$n_A = \frac{P_0V_0}{RT_0}$

$n_B = \frac{10P_0V_0}{RT_0} = 10n_A$

$m_B = 10m_A = 10 \times 100gm = 1kg$

Again for gas is A, $\Delta Q = \Delta U + \Delta W$

$\Rightarrow 0 = \Delta U + \Delta W$

$\Rightarrow 0 = \Delta U + \left\{\frac{P_0V_0 - 8P_0 \times \frac{V_0}{4}}{(\gamma - 1)}\right\}$

$\Rightarrow \Delta U = 2P_0V_0 = 2 \times 10^5 \times 10^{-2} = 2000J$

$F = 8P_0a = 8 \times 10^5 \times 10^{-2} = 8000N$

9. C

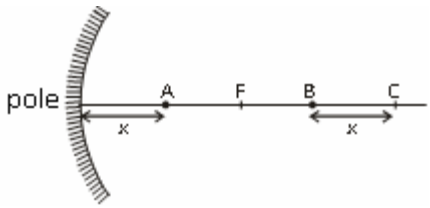
Sol. The image distance of A is

$V_A = -\frac{(f-x)}{xf} (\because x<t)$

The image of B is $V_B = -\frac{[(2f-x)-f]}{(2f-x)-x}$

Solving $|V_A| + |V_B| = 4f$ we get $x = \frac{f}{2}$

$\therefore$ required time $t = \frac{x}{u} = \frac{f}{2u}$



10. B

SECTION – D

11. 00001.00

Sol: at plane surface image of object is $=-\frac{xR}{12}\times 4$ distance at current surface

$u=-\left[\frac{xR}{12}4+R\right]$

$v=\infty$ ROC=-R

$$\frac{\mu_r-\mu_i}{R}=\frac{\mu_r}{v}-\frac{\mu_i}{u}$$

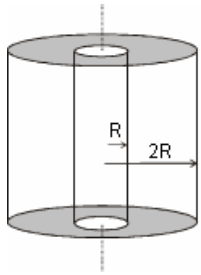
$$\frac{1-4}{-R}=\frac{-4}{\left[\frac{xR4}{12}+R\right]}$$

$\Rightarrow x=1$

12. 00004.00

Sol. $H=-K.2\pi r/\frac{dT}{dr}$

$$\int_{R_1}^{R_2}\frac{Hdr}{2\pi rl}=-K\int_{T_1}^{T_2}dT\Rightarrow H=\frac{2\pi lk\left(T_1-T_2\right)}{\ln\frac{R_2}{R_1}}H_i=H_f$$



13. 00002.00

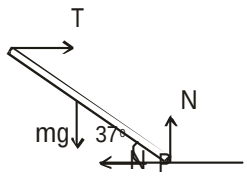
Sol. $\frac{dN_A}{dt}=-\lambda_1 N_A,\frac{dN_B}{dt}=2\lambda_1 N_A-\lambda_2 N_B$

$N_B=\text{maximum}\Rightarrow\frac{dN_B}{dt}=0$

$\Rightarrow 2\lambda_1 N_A=\lambda_2 N_{B_{\max}}\Rightarrow N_{B_{\max}}=\frac{2\lambda_1}{\lambda_2}N_A\Rightarrow N_{B_{\max}}=\frac{2\lambda_1}{\lambda_2}N_0e^{-\lambda_1 t}=2$

14. 00003.00

Sol. T the FBD of any one rod is



$T=\mu N--------(1)$

$mg=N--------(2)$

Taking torque of any one rod is

$mg\frac{L}{g}\cos 37^o=TL\sin 37^o$

$T=\frac{mg}{2}\frac{4}{3}=\frac{2mg}{3}$

$\frac{2mg}{3}=\mu mg\Rightarrow\mu=\frac{2}{3}$

15. 00006.00

Sol. $\frac{1}{f_L}=\left(\frac{3}{2}-1\right)\left(\frac{1}{20}-\frac{1}{20}\right)$

$\Rightarrow F_L=20cm$

$\frac{1}{f}=\frac{1}{-10}-\frac{2}{20}=-\frac{1}{5}$

$\frac{1}{v}+\frac{1}{-30}=\frac{1}{-5}\Rightarrow\frac{1}{v}=\frac{1}{30}-\frac{1}{5}$

For lens $\frac{1}{v}-\frac{1}{-30}=\frac{1}{20}$

16. 00025.00

Sol. $\overline{PQ}=\sqrt{15^2+20^2}=25m$

$25=\frac{V_P^2\sin 2(45)}{g}\Rightarrow V_P=5\sqrt{10}\text{m/s}$

$\therefore V_A^2\sin^2\theta-2\times10\times12.5=\left(5\sqrt{5}\right)^2\Rightarrow V_A\sin\theta=5\sqrt{15}\text{m/s}$

$V_A\cos\theta=V_P\cos45^o=5\sqrt{5}\text{m/s}\,\text{and}\,\theta=60^o$

$\therefore\overline{AB}=\frac{V_A^2\sin2\theta}{g}=25\sqrt{3}$

17. 00030.00

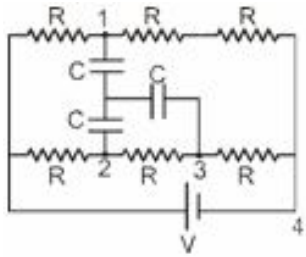
Sol. Consider equation of torque about C

$\left[mg\frac{4R}{3\pi}\right]=\frac{mR^2}{2}\alpha$

$\alpha=\frac{8g}{3\pi R}=\frac{8\pi g}{3\pi^2R}=\frac{8\pi g}{30R}$

18. 00018.00

Sol. No current passes through capacitors in steady state. Assume potential at point '4' to be zero.

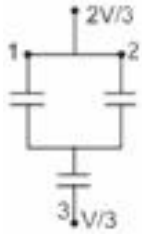


Then points '1' and '2' are at same potential $\frac{2V}{3}$

Hence $C_1\text{and}\,C_2$ can be taken in parallel

The potential at point 3 is $\frac{V}{3}$

$\therefore$ Equivalent circuit of all three capacitors is shown Hence potential difference across capacitor C_3 is



$=\frac{2C}{2C+C}\times\left(\frac{2V}{3}-\frac{V}{3}\right)=\frac{2V}{9}$

Chemistry

PART – II

SECTION – A

19. CD

Sol. It is polymerisation reaction. The chain termination occurs after only isobutylene molecule has been added. This is due to the high concentration of water, the intermediate carbocation loses a proton to water it can react with another alkene molecules.
Hence (C, D) is correct.

20. BD

Sol. The above reaction represent the case of Ar_{SN} (aromatic nucleophilic substitution) which occurs through the formation of benzyne. For the formation of benzyne intermediate, it is essential that an H should be available in on ortho position to the halogen.
Hence (B, D) is correct.

21. AB

Sol. These copolymers are formed in steps and hence step growth polymer

22. AB

Sol. Bronze contains Cu = 80%, Zn = 10%, Sn = 10% whereas brass contains Cu = 80%, Zn = 20%.
Gun metal and type metal do not contain Zn.
Hence, (A) and (B) are the correct answers.

23. AD

Sol. $\underset{\text{Lewis base}}{XeF_2} + \underset{\text{Lewis acid}}{2SbF_5} \rightarrow XeF_2 \cdot 2SbF_5$

$XeF_4 + SbF_5 \rightarrow [XeF_3]^+ [SbF_6]^-$

$XePtF_6$ was first real compound of Xe.

24. ACD

Sol. On increasing the temperature k value increases hence rate of reaction increases thus $t_{1/2}$ decreases.
Hence choice (B) is correct while all others are incorrect.

25. C

Sol. $\log K = \dfrac{nE^\circ}{0.0591} = \dfrac{2 \times 1.23\ V}{0.0591} = 41.62$

$\therefore\ K = 4.210 \times 10^{41}$

26. C

Sol. $\text{Efficiency} = \dfrac{\Delta G}{\Delta H} = \dfrac{-237.39}{-241.00} = 98.5\%$

27. C

Sol. For n^{th} order reaction

$$k_n t = \frac{1}{(n-1)}\left[\frac{1}{a^{n-1}} - \frac{1}{a_0^{n-1}}\right] \quad \ldots(i)$$

$$\text{Hence } k_n t_{1/2} = \frac{1}{n-1}\left[\left(\frac{2}{a_0}\right)^{n-1} - \left(\frac{1}{a_0}\right)^{n-1}\right] \quad \ldots(ii)$$

as $a = \frac{a_0}{2}$

At $t_{3/4}$

$$k_n t_{3/4} = \frac{1}{n-1}\left[\left(\frac{4}{3a_0}\right)^{n-1} - \left(\frac{1}{a_0}\right)^{n-1}\right] \quad \ldots(iii)$$

$a = \frac{3a_0}{4}$ at $t_{3/4}$

From equations (ii) and (iii)

$$\frac{t_{1/2}}{t_{3/4}} = \frac{2^{n-1}-1}{\left(\frac{4}{3}\right)^{n-1}-1}$$

Hence $\frac{t_{1/2}}{t_{3/4}}$ is a function of n.

28. C

Sol. $t_{1/2} \propto \frac{1}{a_0^{n-1}}$

$$\frac{t_{1/2(1)}}{t_{1/2(2)}} = \frac{\left(a_{o(2)}\right)^{n-1}}{\left(a_{0(1)}\right)^{n-1}}$$

$$\left(\frac{4280}{885}\right) = \left(\frac{1.1}{0.5}\right)^{n-1} = \left(\frac{11}{5}\right)^{n-1}$$

Solving we get n = 3

Hence order = 3

SECTION – D

29. 00512.035

Sol: 32. $\frac{P_B}{P_C} = \frac{2\times10^{-3}}{3\times10^{-2}} = \frac{1}{15}$

$\Rightarrow P_C = 15P_B$

$P_0 = 40 + 15 \times 40 = 640$

$(K_1 + K_2) = \frac{2.303}{60}\log\frac{640}{640-p}$

$\therefore P = 546.17 = P_B^0 + P_C = 16P_B$

$\therefore P_B = 34.135$

$P_C = 546.17 - 34.135$

$= 512.035$ mm of Hg.

30. 00937.30

Sol. Let the electron be boosted to n^{th} level, then

Number of spectral lines = $\sum\Delta n = \sum n - 1 = 15$. Given,

Or $\frac{1}{2}[n-1]\times n = 15$

$\therefore$ n = 6

Thus electron is boosted upto 6th energy level from ground state. Therefore

$$\frac{1}{\lambda} = R_H\left[\frac{1}{n^2} - \frac{1}{1^2}\right] = 109737 \times \frac{35}{36}$$

$\lambda = 9.373 \times 10^{-6}$ cm = 937.3 Å

31. 00005.92

Sol. Eq. of metal $= \frac{i.t}{96500}$ (i = 10 ampere, t = 330 sec)

$$\frac{1.95}{E} = \frac{10 \times 330}{96500}$$

$E_{metal} = 57.0$

At. wt. of metal = 57.0×2 (metal is bivalent as salt is MF_2)
= 114.0

Also, if $w_{cu} = 1.95$ g, then

Eq. of Cu $= \frac{i.t}{96500}$

or $\frac{1.95}{63.6/2} = \frac{i.t}{96500}$

$\therefore$ i.t = Q = 5917.45 coulomb = 5.92 kC

32. 00000.37

Sol. For a given aqueous solution

$\Delta T_b = K_b' \times$ molality

$\Delta T_f = K_f' \times$ molality

$$\therefore \quad \frac{\Delta T_b}{\Delta T_f} = \frac{K_b'}{K_f'} = \frac{RT_b^2}{1000\ell_v} \times \frac{1000\ell_f}{RT_f^2}$$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{T_b^2 \times \ell_f}{T_r^{'2} \times \ell_v}$$

$\because$ $T_b = 100 + 273 = 373$ K

$T_f = 0 + 273 = 273$ K

$\ell_f = 80$ cal g^{-1};

$\ell_v = 540$ cal g^{-1}

$$\therefore \quad \frac{0.1}{\Delta T_f} = \frac{373 \times 373 \times 80}{273 \times 273 \times 540}$$

$\therefore \quad \Delta T_f = 0.362$

$\therefore \quad T_f = 0.0 - 0.362 = -0.362^oC$

$|T_f| = 0.362^oC$

33. 00622.40

Sol. Degree of dissociation $(\alpha) = \frac{\wedge_m}{\wedge_m^o}$

$$\Rightarrow \wedge_m = \wedge_m^o \times \alpha$$

$= \wedge_m^o \times \sqrt{\frac{K_a}{C}}$ (by Ostwald dilution law)

$= 390.5 \times \sqrt{\frac{1.8 \times 10^{-5}}{0.04}} = 8.82$ ohm^{-1} cm^2 mol^{-1}

$$\text{Conductivity} = \frac{\text{molarity} \times \wedge_m}{1000}$$

$$= 3.31 \times 10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$$

$$\text{Also conductivity} = \text{Cell constant} \times \frac{1}{\text{Resistance}}$$

$$\Rightarrow \text{Resistance} = \frac{\text{Cell constant}}{\text{Conductivity}}$$

$$= \frac{0.206}{3.31 \times 10^{-2}} = 622.36 \text{ ohm} = 622.4\ \Omega$$

34. 00104.50

Sol. $-\Delta G^0 = 2.303\ RT \log_{10} K$

$\Delta G^0 = \Delta H^0 - T\Delta S^0$

$$\therefore \log K = -\frac{\Delta H^0}{2.303RT} + \frac{\Delta S^0}{2.303 \times R} \quad \text{... (iii)}$$

From equation (i) & (ii)

$$\log \frac{K_f}{K_b} = 5.457 - \frac{5800}{T}$$

$$\Rightarrow \log K = 5.4857 - \frac{5800}{T}$$

$$\log K = -\frac{5800}{T} + 5.457 \quad \text{....(iv)}$$

From equation (iii) & (iv)

$\Delta S^0 = 104.48 JK^{-1}$ = 104.5 JK^{-1}

35. 00675.00

Sol. Let n moles of gas be present

$$\therefore \quad n = \frac{PV}{RT} = \frac{1.6 \times 10^6 \times 0.0083}{8.3 \times 300} = 5.33$$

Given, $C_p = \frac{5}{2}R$, $\therefore\ C_v = \frac{5}{2}R - R = \frac{3}{2}R = \frac{3}{2} \times 8.3 = 12.45\ J/mol-K$

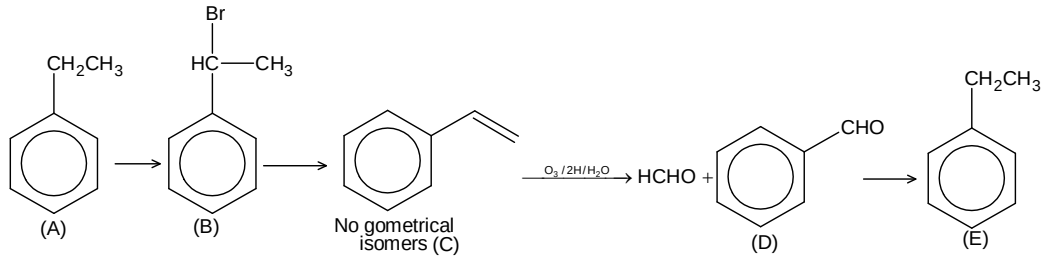
Heat supplied at constant volume = $n \times C_v \times \Delta T$

$\therefore \quad \Delta T = 375\ K$

$\therefore$ Final temperature = 300 + 375 = 675 K

36. 00106.00

Sol.



Also A and E are identical

M. Wt. = 96 + 10 = 106

Number of geometrical isomers = 0

x + y = 106

Mathematics

PART – III

SECTION – A

37. ACD

Sol. $g'(x) = f'(\tan x)\sec^2 x - f'(\cot x)\operatorname{cosec}^2 x$

For increasing $g'(x) > 0$

$\because f''(x) < 0 \Rightarrow f'(x)$ is decreasing for

$\tan x < \cot x \ \forall\ x \in \left(\frac{\pi}{2}, \frac{\pi}{4}\right)$

$\therefore f'(\tan x) > f'(\cot x)$

also, $\sec^2 x > \operatorname{cosec}^2 x \ \forall\ x \in \left(\frac{\pi}{2}, \frac{\pi}{4}\right)$

$\therefore g'(x) > 0 \Rightarrow g(x)$ is increasing in $\left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

similarly $g(x)$ is decreasing in $\left(\frac{3\pi}{4}, \pi\right)$

also $g(x)$ has local max at $x = \frac{3\pi}{4}$.

38. ACD

Sol. $\lim\limits_{x\to -4} \frac{\tan \pi x}{x+4} \qquad \left(\frac{0}{0}\right)$

$= \lim\limits_{x\to -4} \frac{\pi \sec^2 \pi x}{1} = \pi$

$\lim\limits_{x\to\infty}\left(1+\frac{1}{x^2}\right)^x = \lim\limits_{x\to\infty} e^{x\cdot\frac{1}{x^2}} = 1$

given limit $= \pi + 1$.

39. BD

Sol. Area A = area of PQRS

$= (b\sin\theta + a\cos\theta)(a\sin\theta + b\cos\theta)$

$= ab + (a^2 + b^2)\sin\theta\cos\theta$

$= ab + \frac{a^2+b^2}{2}\sin 2\theta$

A is maximum when $\sin 2\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$

$A_{max} = ab + \frac{a^2+b^2}{2} = \frac{1}{2}(a+b)^2$.

40. AC

Sol. Here $x^2 - x + 1 = \left(x-\frac{1}{2}\right)^2 + \frac{3}{4} > 0 \ \ \forall\ x \in R$

gives $(x^2 - x + 1)^{x-1} < 1$

taking log both sides

$(x - 1)\log_{10}(x^2 - x + 1) < 0$

Case I: $x - 1 > 0$, $\log_{10}(x^2 - x + 1) < 0$

i.e. $x - 1 > 0$ and $0 < x < 1 \Rightarrow$ no solution.

Case II: $x - 1 < 0$ and $\log_{10}(x^2 - x + 1) > 0$

$\Rightarrow x < 1$ and $x > 1$ or $x < 0$
$\Rightarrow x < 0$.

41. BD

Sol. If n = m then equation becomes $z^m = \bar{z}^m$ and it has infinite solution because any $z \in$ real will satisfy it.
If $n \neq m$, let n > m, then
$z^n = \bar{z}^m$
$\Rightarrow |z|^n = |z|^m$
$|z|^m (|z|^{n-m} - 1) = 0 \Rightarrow |z| = 0$ or $|z| = 1$
$|z| = 0 \Rightarrow z = 0 + i0$
$|z| = 1 \Rightarrow z = e^{i\theta}$
$\Rightarrow e^{(m+n)\theta i} = 1 \Rightarrow z = 1^{1/(m+n)}$
$\Rightarrow$ number of solution is m + n + 1.

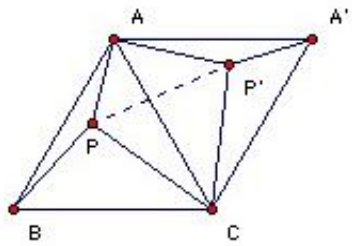
42. ABCD

Sol. Put x = y = z
$\Rightarrow 3|u + v + w|\,|x| = 3|x| \Rightarrow |u + v + w| = 1$
put y = z = 0
$\Rightarrow |ux| + |vx| + |wx| = |x| \Rightarrow |u| + |v| + |w| = 1$
put z = 0 and y = – x
$\Rightarrow |u - v|\,|x| + |v - w|\,|x| + |w - u|\,|x| = 0$
$\Rightarrow u = v = w$.

43. C

44. A

Sol. **(Sol. for Q.43 and Q.44)**
Take the triangle to be ABC and the point P. Let PA = 3, PB = 4, PC = 5. Rotate the triangle and P through 60° about C. Let A go to A' and P to P'. Then CP = CP' and $\angle PCP' = 60^o$, so PCP' is equilateral with side 5. So PAP' is a 3, 4, 5 triangle and hence $\angle PAP' = 90^o$. Hence $\angle PAB + \angle P'AA' = 120^o - 90^o = 30^o$. Hence $\angle APB = 150^o$. Now applying the cosine formula to PAB we have $d^2 = 3^2 + 4^2 - 24\cos 150^o = 25 + 12\sqrt{3}$.



45. C

Sol. $y = \log_{1/2}\left(x - \frac{1}{2}\right) + \log_2 \sqrt{(2x-1)^2}$ But $x > \frac{1}{2}$

$= \log_{1/2}\left(x - \frac{1}{2}\right) + \log_2 (2x - 1)$

y = 1
P ≡ (3, 1).

46. A

Sol. $\overrightarrow{OP} = 3\hat{i} + \hat{j}$
Q ≡ (1, 1) or (2, 1)
$\overrightarrow{OQ} = \hat{i} + \hat{j}$ and $2\hat{i} + \hat{j}$
$\overrightarrow{OP} \cdot \overrightarrow{OQ} = 3 + 1 = 4$ and 6 + 1 = 7.

SECTION – D

47. 00014.60

Sol. Given point is corresponding to t = 2 and $\frac{dy}{dx} = 6t \Rightarrow$ slope of normal at t = 2 is $-\frac{1}{12}$

$\therefore$ equation of normal is $y - 6 = -\frac{1}{12}(x - 1)$

$\Rightarrow a = 73, b = \frac{73}{12}$

$\Rightarrow a + 12b = 146.$

48. 00005.50

Sol. $CT_1 = CT_2 = x$

AB = 20

AC = 11

BC = 13

In ΔABC $\sin A = \frac{3}{5}$ and $\sin B = \frac{33}{65}$ also

$$\frac{r}{\sin A} + \frac{r}{\sin B} = AB$$

$$\Rightarrow r = \frac{11}{2}.$$

49. 00020.00

Sol. $g(x) = x^2 + x^2\int_{-1}^{1} tg(t)dt + x^3\int_{-1}^{1} g(t)dt$

by

$g(x) = x^2(1 + A) + Bx^3$

where $A = \int_{-1}^{1} tg(t)dt$, $B = \int_{-1}^{1} g(t)dt$

$$A = \int_{-1}^{1} t\left[(1+A)t^2 + Bt^2\right]dt = \frac{2B}{5}.$$

Similarly $B = \int_{-1}^{1}\left[(1+A)(1+A)t^2 + Bt^3\right]dt$

$$\Rightarrow B = (A + 1)\frac{2}{3} \Rightarrow 3B = 2\left(1+\frac{2B}{5}\right)$$

$$\Rightarrow A = \frac{4}{11}, B = \frac{10}{11} \Rightarrow g(x) = \frac{15}{11}x^2 + \frac{10}{11}x^3$$

$$\therefore \int_{-1}^{1}\left[g(x) + g(-x)\right]dx = \int_{-1}^{1}\left(\frac{15}{11}x^2 + \frac{10}{11}x^3 + \frac{15}{11}x^2 - \frac{10}{11}x^3\right)dx$$

$$\Rightarrow \int_{-1}^{1}\left[g(x) + g(-x)\right]dx = \frac{30}{11}\int_{-1}^{1} x^2dx = \frac{20}{11} \text{ sq. units.}$$

50. 00090.00

Sol. **Case I:** When we use 6, 7 or 8 at ten thousand place then number of numbers

$= 3 \times {}^4P_4 = 72$

Case II: When we use 5 at ten thousand place and 6, 7 or 8 at thousand place then number of numbers is
$= 1 \times 3 \times {}^3P_3 = 18$
hence the required numbers of numbers is 72 + 18 = 90.

51. 00058.32

Sol. $T_{r+1} = {}^{6561}C_r\,(a^{1/3})^{6561-r}\,(b)^{r/9} = {}^{6561}C_r\,(a)^{(6561-r)/3}\,(b)^{r/9}$

T_{r+1} will be integral if $\frac{r}{9}$ and $\frac{6561-r}{3}$ both are integer.

r = 0, 9, 18, …, 6561.

Number of values of r = $\frac{6561}{9}+1 = 730$.

So, number of irrational terms = 6562 – 730 = 5832.

52. 00672.00

Sol. N = a (111111) is divisible by $7 \times 11 \times 3$.
Hence for N to be a divisible by 924, a = 4 or 8
and a and β are roots of $x^2 - 11x + \lambda = 0$
$\Rightarrow a + \beta = 11$
$\Rightarrow (a, \beta) \equiv (4, 7), (8, 3)$
$\Rightarrow$ possible value of λ = 28, 24
$\Rightarrow$ product of λ = 672.

53. 00100.00

Sol. $2(1 + x^3)^{100} = [a_0 + a_1x + a_2x^2 + \ldots + a_{100}x^{100}]$

$$- \left[\cos\frac{\pi}{2}x + \cos\left(\frac{\pi}{2}x+\frac{\pi}{2}\right) + \ldots + \cos\left(\frac{\pi}{2}x + 100\frac{\pi}{2}\right)\right] \quad \ldots(1)$$

Replace x by – x, we get
$2(1-x^3)^{100} = [a_0 - a_1x + a_2x^2 - \ldots + a_{100}x^{100}]$

$$- \left[\cos\left(-\frac{\pi}{2}x\right) + \cos\left(-\frac{\pi}{2}x+\frac{\pi}{2}\right) + \ldots + \cos\left(-\frac{\pi}{2}x + \frac{100\pi}{2}\right)\right] \quad \ldots(2)$$

Adding (1) and (2), we get
$2(1 + x^3)^{100} + 2(1 - x^3)^{100} = 2(a_0 + a_2x^2 + \ldots + a_{100}x^4)$
Put x = 1, we get $a_0 + a_2 + a_4 + \ldots + a_{100} = 2^{100}$

54. 00040.00

Sol. Let $g(x) = \frac{2007}{x}$.

Let x decreases from k + 1 to k then g(x) increases from $\frac{2007}{k+1}$ to $\frac{2007}{k}$ $\Rightarrow$ The increase in the value of g(x) is $\frac{2007}{k(k+1)}$.

The increase is less than 1 if k(k + 1) > 2007
$\Rightarrow$ k = 45, 46, …, 2007.
$\Rightarrow$ g(x) does not miss any value as x decrease from 2007, 2006, … till 45.

Also $\left[\frac{2007}{45}\right] = 44$, $\left[\frac{2007}{44}\right] = 45$, $\left[\frac{2007}{43}\right] = 46$, $\left[\frac{2007}{42}\right] = 47$, $\left[\frac{2007}{41}\right] = 48$ and $\left[\frac{2007}{40}\right] = 50$

$\Rightarrow \left[\frac{2007}{x}\right]$ Can never be equal to 49

$\Rightarrow$ The maximum value of k is 40.